```
devasc@labvm:~/labs/devnet-src/school-library$ ifconfig --help
Usage:
  ifconfig [-a] [-v] [-s] <interface> [[<AF>] <address>]
  [add <address>[/<prefixlen>]]
  [del <address>[/<prefixlen>]]
  [[-]broadcast [<address>]]  [[-]pointopoint [<address>]]
  [netmask <address>]  [dstaddr <address>]  [tunnel <address>]
  [outfill <NN>] [keepalive <NN>]
  [hw <HW> <address>]  [mtu <NN>]
  [[-]trailers]  [[-]arp]  [[-]allmulti]
  [multicast]  [[-]promisc]
  [mem_start <NN>]  [io_addr <NN>]  [irq <NN>]  [media <type>]
  [txqueuelen <NN>]
  [[-]dynamic]
  [up|down] ...

  <HW>=Hardware Type.
  List of possible hardware types:
    loop (Local Loopback) slip (Serial Line IP) cslip (VJ Serial Line IP)
    slip6 (6-bit Serial Line IP) cslip6 (VJ 6-bit Serial Line IP) adaptive (Adaptive Serial Line IP)
    ash (Ash) ether (Ethernet) ax25 (AMPR AX.25)
    netrom (AMPR NET/ROM) rose (AMPR ROSE) tunnel (IPIP Tunnel)
    ppp (Point-to-Point Protocol) hdlc ((Cisco)-HDLC) lapb (LAPB)
    arcnet (ARCnet) dlci (Frame Relay DLCI) frad (Frame Relay Access Device)
    sit (IPv6-in-IPv4) fddi (Fiber Distributed Data Interface) hippi (HIPPI)
    irda (IrLAP) ec (Econet) x25 (generic X.25)
    eui64 (Generic EUI-64)
  <AF>=Address family. Default: inet
  List of possible address families:
    unix (UNIX Domain) inet (DARPA Internet) inet6 (IPv6)
    ax25 (AMPR AX.25) netrom (AMPR NET/ROM) rose (AMPR ROSE)
    ipx (Novell IPX) ddp (Appletalk DDP) ec (Econet)
    ash (Ash) x25 (CCITT X.25)
```

```
devasc@labvm:~/labs/devnet-src/school-library$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: ens32: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:82:50:77 brd ff:ff:ff:ff:ff:ff
    inet 10.124.34.126/24 metric 100 brd 10.124.34.255 scope global dynamic ens32
       valid_lft 3572sec preferred_lft 3572sec
    inet6 fe80::20c:29ff:fe82:5077/64 scope link
       valid_lft forever preferred_lft forever
3: dummy0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UNKNOWN group default qlen 1000
    link/ether e6:71:9f:99:ba:b9 brd ff:ff:ff:ff:ff:ff
    inet 192.0.2.1/32 scope global dummy0
       valid_lft forever preferred_lft forever
    inet 192.0.2.2/32 scope global dummy0
       valid_lft forever preferred_lft forever
    inet 192.0.2.3/32 scope global dummy0
       valid_lft forever preferred_lft forever
    inet 192.0.2.4/32 scope global dummy0
       valid_lft forever preferred_lft forever
    inet 192.0.2.5/32 scope global dummy0
       valid_lft forever preferred_lft forever
    inet6 fe80::e471:9fff:fe99:bab9/64 scope link
       valid_lft forever preferred_lft forever
4: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
    link/ether 02:42:d8:86:6b:6e brd ff:ff:ff:ff:ff:ff
```

```
devasc@labvm:~/labs/devnet-src/school-library$ ping -help

Usage
  ping [options] <destination>

Options:
  <destination>      dns name or ip address
  -a                 use audible ping
  -A                 use adaptive ping
  -B                 sticky source address
  -c <count>         stop after <count> replies
  -D                 print timestamps
  -d                 use SO_DEBUG socket option
  -f                 flood ping
  -h                 print help and exit
  -I <interface>     either interface name or address
  -i <interval>      seconds between sending each packet
  -L                 suppress loopback of multicast packets
  -l <preload>       send <preload> number of packages while waiting replies
  -m <mark>          tag the packets going out
  -M <pmtud opt>     define mtu discovery, can be one of <do|dont|want>
  -n                 no dns name resolution
  -O                 report outstanding replies
  -p <pattern>       contents of padding byte
  -q                 quiet output
  -Q <tclass>        use quality of service <tclass> bits
  -s <size>          use <size> as number of data bytes to be sent
  -S <size>          use <size> as SO_SNDBUF socket option value
  -t <ttl>           define time to live
  -U                 print user-to-user latency
  -v                 verbose output
  -V                 print version and exit
  -w <deadline>      reply wait <deadline> in seconds
  -W <timeout>       time to wait for response

IPv4 options:
  -4                 use IPv4
```

```
devasc@labvm:~/labs/devnet-src/school-library$ ping -c 5 www.cisco.com
PING e2867.dsca.akamaiedge.net (23.197.160.120) 56(84) bytes of data.
64 bytes from a23-197-160-120.deploy.static.akamaitechnologies.com (23.197.160.120): icmp_seq=1 ttl=43 time=326 ms
64 bytes from a23-197-160-120.deploy.static.akamaitechnologies.com (23.197.160.120): icmp_seq=2 ttl=43 time=208 ms
64 bytes from a23-197-160-120.deploy.static.akamaitechnologies.com (23.197.160.120): icmp_seq=3 ttl=43 time=168 ms
64 bytes from a23-197-160-120.deploy.static.akamaitechnologies.com (23.197.160.120): icmp_seq=4 ttl=43 time=214 ms
64 bytes from a23-197-160-120.deploy.static.akamaitechnologies.com (23.197.160.120): icmp_seq=5 ttl=43 time=190 ms

--- e2867.dsca.akamaiedge.net ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4007ms
rtt min/avg/max/mdev = 167.708/221.072/326.120/54.901 ms
devasc@labvm:~/labs/devnet-src/school-library$ traceroute --help
Usage: traceroute [OPTION...] HOST
Print the route packets trace to network host.

  -f, --first-hop=NUM        set initial hop distance, i.e., time-to-live
  -g, --gateways=GATES       list of gateways for loose source routing
  -I, --icmp                 use ICMP ECHO as probe
  -m, --max-hop=NUM          set maximal hop count (default: 64)
  -M, --type=METHOD          use METHOD (`icmp' or `udp') for traceroute
                             operations, defaulting to `udp'
  -p, --port=PORT            use destination PORT port (default: 33434)
  -q, --tries=NUM            send NUM probe packets per hop (default: 3)
      --resolve-hostnames    resolve hostnames
  -t, --tos=NUM              set type of service (TOS) to NUM
  -w, --wait=NUM             wait NUM seconds for response (default: 3)
  -?, --help                 give this help list
      --usage                give a short usage message
  -V, --version              print program version

Mandatory or optional arguments to long options are also mandatory or optional
for any corresponding short options.

Report bugs to <bug-inetutils@gnu.org>.
```

```
devasc@labvm:~/labs/devnet-src/school-library$ nslookup www.cisco.com
Server:         127.0.0.53
Address:        127.0.0.53#53

Non-authoritative answer:
www.cisco.com   canonical name = www.cisco.com.akadns.net.
www.cisco.com.akadns.net        canonical name = wwwds.cisco.com.edgekey.net.
wwwds.cisco.com.edgekey.net     canonical name = e2867.dsca.akamaiedge.net.
Name:   e2867.dsca.akamaiedge.net
Address: 23.197.160.120
Name:   e2867.dsca.akamaiedge.net
Address: 2600:140f:4:68c::b33
Name:   e2867.dsca.akamaiedge.net
Address: 2600:140f:4:693::b33
```

```
devasc@labvm:~/labs/devnet-src/school-library$ nslookup 8.8.8.8
8.8.8.8.in-addr.arpa    name = dns.google.

Authoritative answers can be found from:

devasc@labvm:~/labs/devnet-src/school-library$ nslookup www.cisco.com 8.8.8.8
Server:         8.8.8.8
Address:        8.8.8.8#53

Non-authoritative answer:
www.cisco.com   canonical name = www.cisco.com.akadns.net.
www.cisco.com.akadns.net        canonical name = wwwds.cisco.com.edgekey.net.
wwwds.cisco.com.edgekey.net     canonical name = e2867.dsca.akamaiedge.net.
Name:   e2867.dsca.akamaiedge.net
Address: 23.197.160.120
Name:   e2867.dsca.akamaiedge.net
Address: 2600:140f:4:693::b33
Name:   e2867.dsca.akamaiedge.net
Address: 2600:140f:4:68c::b33

devasc@labvm:~/labs/devnet-src/school-library$ tracert www.netacad.com

Command 'tracert' not found, did you mean:

  command 'tracert6' from deb ndisc6 (1.0.4-1ubuntu1)
  command 'tracer' from deb pvm-dev (3.4.6-2build2)

Try: sudo apt install <deb name>

devasc@labvm:~/labs/devnet-src/school-library$
```

```
C:\Windows\System32>tracert www.netacad.com

Трассировка маршрута к d1h6v4iwmfkzng.cloudfront.net [18.165.140.116]
с максимальным числом прыжков 30:

  1     *        *        *     Превышен интервал ожидания для запроса.
  2    32 ms    21 ms    30 ms  10.250.28.158
  3     *        *        *     Превышен интервал ожидания для запроса.
  4   166 ms    57 ms    50 ms  10.240.30.201
  5    47 ms    26 ms    33 ms  10.240.30.82
  6    57 ms     *        *     109-71-224-205.megacom.kg [109.71.224.205]
  7    48 ms    30 ms    23 ms  109-71-224-161.megacom.kg [109.71.224.161]
  8    20 ms    25 ms    25 ms  mx-bish-2.elcat.kg [94.143.195.68]
  9    84 ms    78 ms    56 ms  ae16.rt1.int.stv.ovl.retn.net [87.245.238.6]
 10    99 ms    84 ms   150 ms  ae10-32.rt.tnr.hki.fi.retn.net [87.245.234.29]
 11     *        *        *     Превышен интервал ожидания для запроса.
 12     *        *        *     Превышен интервал ожидания для запроса.
 13     *        *        *     Превышен интервал ожидания для запроса.
 14     *        *        *     Превышен интервал ожидания для запроса.
 15   229 ms   295 ms   109 ms  server-18-165-140-116.hel51.r.cloudfront.net [18.165.140.116]

Трассировка завершена.

C:\Windows\System32>
```